

TAKE-HOME ASSIGNMENT

Daily Food Ordering Agent Skill

Inference.ai • Product & Strategy Intern

Role	AI Product & Strategy Intern, Inference.ai
Deadline	48 hours from receipt of this document
Deliverable	SKILL.md specification + working open-source implementation
Constraint	Open-source tooling only. Do not use Claude Code CLI.
Submission	GitHub repository link or compressed archive via your recruiter

Overview

At Inference.ai, we build AI agents that take meaningful action on behalf of users, reducing friction on repetitive tasks so people can focus on work that actually requires them. This exercise is designed to reflect how we think about agent design: not just whether an agent completes a task, but how it reasons, degrades gracefully under failure, and earns user trust over time.

Your objective is to design and implement a **skill**: a self-contained capability module consisting of a structured specification (SKILL.md) and working code that enables an AI agent to automatically order food for a user once per day on a configurable schedule.

Deliverables

Your submission should include the following:

1. SKILL.md: Agent Specification

- Frontmatter including skill name and a description of when and how the agent should invoke it
- Ordered instructions the agent follows when executing a food order
- A user-facing configuration schema defining preference inputs
- Documented error handling and fallback behavior

2. Live Presentation

- A walkthrough of your SKILL.md and the decisions behind it
 - How the agent handles edge cases and user preferences
 - What you would build or refine next given more time
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Constraints

- Open-source tooling only. Do not use Claude Code CLI.
 - The skill must be designed to work with any LLM backend, or none at all
 - 48 hours is a tight window. Prioritize depth and clarity over breadth.
 - A well-reasoned, focused SKILL.md is preferable to one that tries to cover everything.
 - Any scope decisions or assumptions should be noted so you can speak to them in your presentation
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Reference: User Preference Schema

The schema below is provided as a starting point only. You are encouraged to redesign it entirely. The key requirement is that it gives the agent enough signal to make consistent, personalized decisions without requiring user input on each run.

```
# user_preferences.yaml
order_time: "12:00" # Daily trigger time (user local)
cuisine_preferences:
  - Japanese
  - Mediterranean
dietary_restrictions:
  - no shellfish
budget_max_usd: 20
confirmation_required: true # Prompt user before placing order
fallback_restaurant: "Sweetgreen, location_id: 4421"
```

Submission

- Share a GitHub repository link, or send a compressed archive to your recruiter
- Be prepared to walk through your SKILL.md and present your approach
- Deadline: 48 hours from receipt of this document
- Questions are welcome. Contact your recruiter rather than making assumptions

What to Submit

File / Folder	Purpose
SKILL.md	Agent skill specification (primary deliverable)
user_preferences.yaml	Sample user configuration
README.md	Assumptions and what you would build next